Cross-Representation Disagreement as a Diagnostic Signal for LLM Reasoning Errors

Anonymous

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Abstract

When a large language model formalizes a reasoning problem into two distinct executable representations—first-order logic with Z3 verification and executable Python—disagreements between the outputs serve as a high-precision diagnostic signal. On three benchmarks (FOLIO, LogiQA, MATH Level 5) and two models (Qwen3-32B, DeepSeek V4 Pro), we show that cross-representation disagreement (CRD) detects errors with 76–87% diagnostic precision at 28–42% disagreement rates, using only 40% of the token cost of $k = 5$ self-consistency. We contribute a four-type error taxonomy validated across models (Cohen's $\kappa = 0.67$), reveal a GT-label-dependent repair effect where diagnosis helps for uncertain cases (+22.1pp on GT=Unknown) but hurts for true cases (-11.2pp overcorrection), and demonstrate that asymmetry direction reverses across domains—FOL-favored in logic (5.6:1), Code-favored in math (1.8:1)—indicating that disagreement patterns reflect domain-specific representation strengths rather than fixed verifier superiority.

1 Introduction

Neurosymbolic reasoning pipelines increasingly rely on LLMs to translate natural language into formal executable representations Pan et al. [2023], Ye et al. [2024]. The dominant failure mode is not the symbolic solver but the LLM's formalization: subtle corruptions introduced during translation from natural language to formal representation Olausson et al. [2023]. Detecting these errors is difficult because the formal output is syntactically valid and the solver returns a definitive answer, masking upstream formalization errors.

Current approaches to this problem fall into three categories: (1) *selection* methods that choose among representations based on confidence Li et al. [2024], Sun et al. [2024], (2) *within-representation verification* that checks consistency within a single paradigm Weng et al. [2023], Kang et al. [2025], Zhong et al. [2025], and (3) *cross-representation filtering* that discards samples where two formalizations disagree Liang et al. [2024], PROVE [2024]. None of these exploit the diagnostic information *within* the disagreement pattern itself—what type of error caused the disagreement, which representation is likely wrong, and how to repair it.

We present **Cross-Representation Disagreement Diagnosis (CRD)**, an analysis framework and lightweight repair method that extends cross-representation disagreement from a binary filtering signal to a structured diagnostic tool. Given a reasoning problem, we generate two executable formalizations (FOL+Z3 and Python), execute both, and when they disagree, classify the error type, localize the faulty representation via stability testing, and optionally perform diagnosis-guided repair. Our three claims:

1. **Disagreement as high-precision diagnostic signal.** Cross-representation disagreement occurs at 28–42% rates across benchmarks and is diagnostically informative with 76–87% precision—substantially higher than $k = 5$ single-representation self-consistency (71–79%) at 40% of the token cost.

2. **Structured error taxonomy with prescriptive value.** Disagreement patterns enable classification into four error types (logic_error, negation_error, exec_crash, quantifier_error). This taxonomy is

stable across models ($\kappa = 0.67$) and has prescriptive value: taxonomy-guided repair outperforms fixed-strategy repair by +5.0pp ($p = 0.042$).

3. **GT-label-dependent repair effect.** Diagnosis-guided repair exhibits a striking asymmetry: it substantially helps for GT=Unknown (+22.1pp) and GT=False (+10.4pp) cases but *hurts* for GT=True cases (-11.2pp). The aggregate masks a three-way split that is the actual finding. We characterize this as a fundamental boundary of cross-representation diagnosis.

Our primary contribution is the systematic analysis of cross-representation disagreement as a diagnostic signal across domains and models. The repair framework demonstrates downstream utility but is secondary to the analytical findings. We release all experimental data and analysis code.

2 Related Work

Neurosymbolic Reasoning. Logic-LM Pan et al. [2023] and LINC Olausson et al. [2023] translate NL to FOL for symbolic solving. SatLM Ye et al. [2024] uses SAT solvers. These methods use a single representation and cannot detect formalization errors through cross-representation comparison.

Cross-Representation Filtering. Liang et al. Liang et al. [2024] showed CoT-PoT disagreement is informative for filtering but stopped at discarding disagreements—no error categorization, no repair. PROVE PROVE [2024] similarly filters but does not diagnose. Our work extends filtering to structured diagnosis.

Within-Representation Verification. VeriCoT Weng et al. [2023] and VERGE Kang et al. [2025] verify reasoning chains within a single paradigm. Roundtrip Verification Zhong et al. [2025] demonstrated diagnosis-guided repair within one representation on a narrow domain (150 traffic rules). We extend to cross-representation diagnosis across multiple domains.

Self-Consistency and Ensemble Methods. Self-consistency Wang et al. [2023] uses k samples from the same prompt for majority voting. Multi-agent debate Du et al. [2023], Liang et al. [2023] and LEVER Ni et al. [2023] leverage multiple generation attempts. A key distinction: within-representation sampling captures generation variance, while CRD captures systematic paradigm-level differences between formal logic and imperative code.

Self-Debugging and Refinement. Self-Debug Chen et al. [2023] and Self-Refine Madaan et al. [2023] use execution feedback for iterative correction within a single representation. Faithful CoT Lyu et al. [2023] ensures reasoning chains are executable. These operate within one paradigm; CRD uses cross-paradigm disagreement as the diagnostic trigger.

Adaptive Selection. HybridMind Li et al. [2024] and adaptive selection methods Sun et al. [2024] choose between representations based on problem features. They select but do not diagnose or repair. CRD is complementary—it diagnoses *after* both representations are generated.

Correlated Failures. Denisov-Blanch Denisov-Blanch [2026] raised concerns about correlated failures in ensemble methods. We explicitly quantify the false agreement rate (12.3%) and show it varies by GT label, with the lowest correlation ($\phi = 0.098$) on the subset where CRD is most effective (GT=Unknown).

3 Method

3.1 Overview

Given a reasoning problem x with ground-truth label $y \in \{\text{True}, \text{False}, \text{Unknown}\}$, CRD proceeds in four stages:

1. **Dual formalization.** Prompt the LLM to produce two executable formalizations: $f_{\text{FOL}}(x)$ (first-order logic formula verified by Z3) and $f_{\text{Code}}(x)$ (Python program with execution).
2. **Disagreement detection.** Execute both formalizations and compare outputs. If $f_{\text{FOL}}(x) \neq f_{\text{Code}}(x)$, the sample enters the diagnostic pipeline.
3. **Error classification and localization.** Classify the disagreement into one of four error types using a rule-based taxonomy. Apply a stability test (re-execution with perturbed prompts) to localize the faulty representation.
4. **Diagnosis-guided repair** (optional). Generate a repair prompt incorporating the diagnosis (error type, suspected faulty side, disagreement pattern) and re-generate the faulty formalization.

For mathematical reasoning, we instantiate the dual representations as SymPy symbolic computation and Python numerical computation, maintaining the cross-executable-paradigm structure.

3.2 Error Taxonomy

We define four error types based on analysis of disagreement patterns:

- **logic_error**: Incorrect predicate definitions, flawed inference chains, or wrong logical connectives in the formalization.
- **negation_error**: Incorrect handling of negation—either missing negation or spurious double negation.
- **exec_crash**: Runtime errors in the executable path (syntax errors, undefined variables, solver timeouts).
- **quantifier_error**: Incorrect scope or type of quantifiers ($\forall$ vs. $\exists$) in FOL formalization.

The taxonomy is deliberately rule-based for interpretability. We validate its cross-model stability in Section 4.10.

3.3 Stability-Based Localization

To determine which representation contains the error, we apply a stability test: re-run each formalization $k=3$ times with temperature perturbation. If one representation produces consistent outputs while the other fluctuates, the unstable side is localized as the error source. When neither side is clearly unstable, the localization returns "ambiguous."

This yields a high-precision / low-recall localization: when the stability test commits (53% of diagnostic cases on FOLIO), accuracy is 94.6%. The 47% ambiguous rate is primarily driven by GT=Unknown samples where Code overclaiming creates systematically unstable predictions.

3.4 Multi-Round Repair

Repair proceeds for up to 3 rounds. Each round generates a repair prompt incorporating: (1) the error type from the taxonomy, (2) the localized faulty side (if available), (3) the outputs from both representations. The repair prompt is type-specific: logic errors receive prompts targeting predicate and inference chain review; negation errors receive prompts checking polarity; exec_crash triggers re-generation; quantifier errors receive prompts reviewing quantifier scope.

Metric	Estimate	95% CI	Gate
C1 (Disagree Rate)	42.16%	[35.3, 49.0]	$\geq$15%
C2 (Diagnostic Rate)	87.21%	[81.3, 94.7]	$\geq$75%
C3 (Error Types)	4	—	$\geq$3
C4 (Repair Δ)	+4.65pp	[$-$8.8, +18.1]	n.s.
C5 (Committed Acc)	94.6%	[86.1, 100]	—
C5 (Coverage)	53.0%	[41.0, 65.1]	—

Table 1: FOLIO C1–C5 metrics with bootstrap 95% CI (Qwen3-32B, $n = 204$). C2 lower bound 81.3% exceeds the 75% threshold, confirming high diagnostic precision.

Error Type	Count	Proportion
logic_error	52	25.5%
negation_error	22	10.8%
exec_crash	11	5.4%
quantifier_error	4	2.0%

Table 2: Error taxonomy distribution on FOLIO disagreement samples.

3.5 Cost Analysis

CRD requires exactly 2 LLM calls for dual formalization (1 FOL + 1 Code), compared to $k = 5$ calls for self-consistency. The repair pipeline adds 1–3 calls only for disagreeing samples. Overall token cost is approximately $2/5 = 0.40$ of SC-5, with additional diagnosis tokens for the disagreeing subset.

4 Experiments

4.1 Setup

Models. Qwen3-32B (primary), DeepSeek V4 Pro (cross-model validation), Qwen3-8B (scale ablation).

Benchmarks. (1) FOLIO Han et al. [2022]: expert-annotated FOL problems, 204 samples, 3-class (True/False/Unknown). (2) LogiQA Liu et al. [2021]: natural language logical reasoning from Chinese civil service exams, 200 samples, 4-way multiple choice. (3) MATH Level 5 Hendrycks et al. [2021]: competition mathematics, 200 samples, using SymPy symbolic vs. Python numerical as dual representations. (4) GPQA Diamond Rein et al. [2024]: graduate-level science questions, 150 samples (secondary benchmark).

Metrics. C1: disagreement rate. C2: diagnostic rate (fraction of disagreements where at least one side is wrong). C3: number of distinct error types observed. C4: repair delta (accuracy improvement over blind baseline). C5: localization accuracy (committed accuracy and coverage).

4.2 FOLIO Core Analysis

Table 1 presents the core metrics on FOLIO with Qwen3-32B (204 samples).

The disagreement rate of 42.16% indicates that a substantial fraction of problems elicit different answers from FOL+Z3 vs. Python execution. Of these disagreements, 87.21% are diagnostically informative (at least one side is wrong). The error taxonomy identifies four types, dominated by logic_error (25.5%), followed by negation_error (10.8%), exec_crash (5.4%), and quantifier_error (2.0%) (Table 2).

The representation asymmetry is striking: FOL is correct in 56 disagreement cases vs. Code correct in 10 and both wrong in 10, yielding a 5.6:1 FOL-favored ratio. The phi coefficient between FOL and Code

GT	n	Targeted	Blind	Δ	Interp.
True	72	63.6%	77.3%	-13.6pp	Overcorrect
False	63	76.7%	70.0%	+6.7pp	Modest help
Unknown	69	67.7%	52.9%	+14.7pp	Diagnosis helps

Table 3: GT-label-stratified repair effect on FOLIO (Round 1). Aggregate non-significance masks a three-way split.

GT Label	n	Rd. 1 Δ	Rd. 3 Δ	Improv.
Unknown	69	+14.7pp	+22.1pp	+7.4pp
False	63	+6.7pp	+10.4pp	+3.7pp
True	72	-13.6pp	-11.2pp	+2.4pp

Table 4: Multi-round repair results on FOLIO. GT=Unknown Δ rises from +14.7 to +22.1pp; GT=True overcorrection partially mitigated but remains negative.

failures is 0.189 ($p = 0.007$), indicating low but statistically significant correlation—the representations fail largely independently, with 75% of errors being single-sided.

4.3 GT-Label-Dependent Repair Effect

The aggregate C4 of +4.65pp (Fisher $p = 0.626$) is not statistically significant. However, stratification by ground-truth label reveals a three-way split (Table 3).

The GT=Unknown subset benefits most from diagnosis because Python's Code path lacks native three-valued logic and tends to overclaim True or False (78.3% overclaiming rate). When CRD diagnosis reveals that FOL returned Unknown, the repair prompt helps the model recognize insufficient evidence. GT=True overcorrection occurs because when Code erroneously outputs False/Unknown, the diagnostic signal triggers the model to move away from True.

4.4 Multi-Round Repair

Table 4 shows how repair delta evolves across three rounds with error-type-specific prompts.

Multi-round repair exhibits diminishing returns (Round 2: 8 additional fixes, Round 3: 3), confirming that type-specific strategies provide incremental value. Table 5 breaks down repair success by error type.

4.5 Cross-Model Ablation

To verify that CRD properties are not specific to Qwen3-32B, we test three conditions on FOLIO (Table 6): (a) Qwen3-32B within-model (MVP), (b) DeepSeek V4 Pro within-model, and (c) DeepSeek FOL + Qwen Code cross-model.

DeepSeek within-model C2 (83.5%) falls within $\pm$8pp of Qwen (87.2%), confirming that diagnostic precision is a structural property of cross-representation disagreement rather than model-specific. The cross-model condition shows higher C1 (47.5%) because different models' formalization biases increase disagreement probability, but slightly lower C2 (80.9%) because some disagreement arises from capability differences rather than formalization errors. The GT-label-dependent repair effect replicates: DeepSeek shows +12.1pp on GT=Unknown and -10.3pp on GT=True.

4.6 Cross-Benchmark Generalization

LogiQA. We first piloted ProofWriter ($n = 30$) but found only 6.7% disagreement rate—synthetic data is too regular for CRD to be informative (both FOL and Code achieve >90% accuracy). We switched to

Error Type	Attempted	Fixed	Success
logic_error	41	28	68.3%
negation_error	18	13	72.0%
exec_crash	11	9	81.8%
quantifier_error	4	2	50.0%

Table 5: Repair success rate by error type (Round 3, FOLIO). exec_crash is easiest to fix; quantifier_error is hardest.

Condition	C1	C2	C2 CI	Asym.
Qwen within	42.2%	87.2%	[81.3, 94.7]	5.6:1 FOL
DeepSeek within	38.7%	83.5%	[74.2, 91.8]	4.2:1 FOL
DS×Qwen cross	47.5%	80.9%	[71.8, 89.1]	3.8:1 FOL

Table 6: Cross-model ablation on FOLIO ($n = 204$). C2 remains above 80% across conditions. Asymmetry direction is consistently FOL-favored.

LogiQA, a natural-language logic benchmark derived from Chinese civil service exams. Table 7 compares LogiQA with FOLIO.

LogiQA C2 of 81.5% (CI: [73.8%, 88.2%]) confirms cross-benchmark stability. The narrower asymmetry (3.2:1 vs. 5.6:1) reflects LogiQA's propositional logic being less demanding than FOLIO's first-order logic, reducing FOL's relative advantage.

ProofWriter pilot failure. The 6.7% pilot disagreement rate on ProofWriter (Table 8) is itself informative: CRD requires natural-language formalization challenges to produce a useful disagreement signal. Synthetic benchmarks with regular structure are well-handled by both representations, limiting CRD's applicability.

MATH Level 5. We extend CRD to mathematical reasoning using SymPy symbolic computation vs. Python numerical computation as dual executable representations (Table 9).

The asymmetry reversal is a key finding (Table 10). In logic, FOL+Z3's formal verification guarantee gives it a systematic advantage. In mathematics, Python's flexible numerical methods are more robust than SymPy's symbolics (SymPy crash rate 12.5% vs. Python 4.0%). This proves that CRD captures *domain-specific representation strengths*, not a fixed verifier hierarchy.

4.7 Correlated Failure Analysis

A natural concern is correlated failures: if both representations fail on the same samples, disagreement underestimates the true error rate. Table 11 quantifies this.

The overall false agreement rate is 12.3%—relatively low, and lowest on GT=Unknown (5.8%) where CRD repair delta is highest. The phi coefficient is 0.189 overall ($p = 0.007$), falling in the range where representations are independent enough to generate complementary signals but not so independent that disagreement is noise. Specification errors (ambiguity in the original problem) account for only 8.3% of cases, confirming that most errors are formalization-specific.

4.8 Selection vs. Repair Ablation

After detecting disagreement and localizing the error side, should one simply *select* the other side, or *repair* the faulty one? Table 12 compares on the GT≠True subset ($n = 132$), where diagnosis is beneficial.

Selection alone provides +11.3pp over blind baseline, demonstrating that CRD's diagnostic value is substantial even without repair. Repair adds +3.5pp on top, with the advantage concentrated on logic_error

Metric	LogiQA	FOLIO	Δ
C1 (Disagree Rate)	34.0%	42.2%	-8.2pp
C2 (Diagnostic Rate)	81.5%	87.2%	-5.7pp
C3 (Error Types)	4	4	—
C5 Committed Acc	91.2%	94.6%	-3.4pp
Asymmetry	3.2:1 FOL	5.6:1 FOL	narrower

Table 7: LogiQA results ($n = 200$, Qwen3-32B) vs. FOLIO reference. C2 remains above 80%; the GT-stratified repair effect replicates.

Metric	Value	Gate	Decision
Disagree Rate	6.7%	$\geq$10%	FAIL
FOL Accuracy	93.3%	—	—
Code Accuracy	90.0%	—	—

Table 8: ProofWriter pilot ($n = 30$). High accuracy on both paths yields insufficient disagreement for CRD.

(+4.9pp) and quantifier_error (+25.0pp)—error types where both sides may be partially correct, making simple selection insufficient.

4.9 Taxonomy-Guided vs. Fixed-Strategy Repair

To test the prescriptive value of the error taxonomy, we compare taxonomy-guided repair (type-specific prompts) against a fixed strategy (uniform re-generation) in Table 13.

The +5.0pp overall advantage is statistically significant (McNemar $p = 0.042$). The taxonomy's prescriptive value is concentrated on errors requiring targeted understanding: logic_error (+6.2pp, prompts targeting predicate and inference chain review) and quantifier_error (+18.0pp, prompts reviewing quantifier scope). exec_crash shows zero difference because re-generation is inherently the correct strategy regardless of diagnosis.

4.10 Cross-Model Taxonomy Validation

To validate that the error taxonomy is not an artifact of a single model's classification, we have DeepSeek V4 Pro independently classify the same 75 diagnostic FOLIO samples (Table 14).

Cohen's $\kappa = 0.67$ (CI: [0.54, 0.80]) reaches "substantial agreement." The rank order of error types is identical across models (logic $>$ negation $>$ crash $>$ quantifier), suggesting the taxonomy captures genuine structural failure modes rather than model-specific artifacts. exec_crash achieves the highest agreement (90.9%) due to its objective verifiability; quantifier_error the lowest (50.0%) due to small sample size ($n = 4$).

4.11 CRD vs. Self-Consistency

We compare single-pass CRD against $k = 5$ self-consistency (majority vote) within each representation (Table 15).

CRD Pareto-dominates $k = 5$ self-consistency: +4.9pp average precision at 40% token cost. The advantage is largest on FOLIO (+7.9pp), where FOL and Python represent maximally different paradigms, and smallest on MATH (+1.2pp), where SymPy and Python numerical are both Python-ecosystem tools with smaller paradigm distance. On FOLIO, the precision gap is statistically significant (McNemar $p = 0.031$). This result supports the thesis that cross-representation disagreement captures paradigm-level differences—fundamentally more informative than within-representation sampling variance.

Metric	MATH	FOLIO	LogiQA
C1	28.5%	42.2%	34.0%
C2	76.0%	87.2%	81.5%
C3	3	4	4
Asymmetry	1.8:1 Code	5.6:1 FOL	3.2:1 FOL
C5 Committed	88.5%	94.6%	91.2%

Table 9: MATH Level 5 results ($n = 200$, Qwen3-32B). Asymmetry direction *reverses*: Python numerical outperforms SymPy symbolic, contrasting with FOL-favored logic benchmarks.

Domain	Rep. A	Rep. B	Favored	Ratio
Logic (FOLIO)	FOL+Z3	Python	FOL	5.6:1
Logic (LogiQA)	FOL+Z3	Python	FOL	3.2:1
Math (MATH)	SymPy	Python	Python	1.8:1

Table 10: Asymmetry direction across domains. The reversal from FOL-favored (logic) to Code-favored (math) reflects representation-task fit.

4.12 Model Scale Ablation

Table 16 compares three models on FOLIO, forming a capability gradient.

Weaker models produce more disagreements (C1: 55.4% for 8B vs. 42.2% for 32B) but with lower diagnostic value (C2: 74.5% vs. 87.2%). The 8B model's C2 of 74.5% approaches the lower bound of usefulness—below some capability threshold, correlated failures (both sides wrong) begin to dominate disagreement, eroding diagnostic precision. The asymmetry ratio compresses from 5.6:1 to 2.8:1 because FOL accuracy degrades faster with model scale (formal logic is more capability-sensitive), reducing the structural advantage.

4.13 Cross-Domain Boundary: GPQA Diamond

To characterize the boundary of CRD's applicability, we test on GPQA Diamond, a graduate-level science reasoning benchmark (Table 17).

The four benchmarks form a clear domain gradient: C2 decreases from 87.2% (pure logic) through 81.5% and 76.0% to 72.7% (science) as the primary error source shifts from formalization errors to domain knowledge gaps. CRD's diagnostic power derives from detecting formalization-specific errors; when the model lacks the domain knowledge to formalize correctly in *any* representation, disagreement provides weaker signal.

4.14 Statistical Significance

Table 18 summarizes significance tests across all benchmarks.

All primary claims at the stratified level reach statistical significance. The non-significant aggregate C4 reinforces our methodological argument: reporting aggregate repair delta masks the three-way split and produces a misleading null result. Bootstrap 95% CIs for C2 across benchmarks are: FOLIO [81.3%, 94.7%], LogiQA [73.8%, 88.2%], MATH [66.1%, 85.2%].

5 Discussion

Paradigm distance modulates CRD effectiveness. The gradient from FOLIO (+7.9pp over SC5) to MATH (+1.2pp) suggests that CRD's advantage over within-representation self-consistency scales with the semantic distance between representations. FOL+Z3 and Python represent fundamentally different reasoning paradigms (declarative vs. imperative); SymPy and Python numerical are both imperative

GT	n	ϕ	False Agr.	Code Overcl.
True	72	0.245	19.4%	N/A
False	63	0.162	11.1%	N/A
Unknown	69	0.098	5.8%	78.3%
Overall	204	0.189	12.3%	—

Table 11: Correlated failure analysis by GT label. GT=Unknown has the lowest failure correlation ($\phi = 0.098$) and highest Code overclaiming (78.3%), explaining why CRD is most effective on this subset.

Method	All	logic	neg.	crash	quant.
Blind	61.4	56.1	66.7	72.7	25.0
Selection	72.7	70.7	77.8	90.9	50.0
Repair (Rd. 3)	76.2	75.6	77.8	90.9	75.0

Table 12: Selection vs. Repair on GT$\neq$True subset (FOLIO, $n = 132$). All values are accuracy (%). Selection already yields +11.3pp over blind; repair adds +3.5pp, concentrated on logic_error and quantifier_error.

and Python-based, limiting paradigm-level diversity. This points to a design principle: maximize representation paradigm distance when deploying CRD.

When NOT to use CRD. ProofWriter's 6.7% disagreement rate and GPQA's 72.7% C2 delineate the boundaries. CRD is most valuable when (1) the task involves natural-language formalization challenges (not synthetic/regular data) and (2) errors arise from the formalization process rather than missing domain knowledge. It provides diminishing returns on tasks where both representations achieve high accuracy or where knowledge gaps dominate.

The GT=True overcorrection boundary. GT=True overcorrection (-11.2pp after 3 rounds) is structural, not strategic. When Code erroneously outputs False/Unknown for a True problem, the diagnostic signal "Code disagrees" pushes the model away from the correct answer. Multi-round repair mitigates this (from -13.6 to -11.2pp) but cannot eliminate it without access to ground truth. Practitioners should apply diagnosis-guided repair selectively, skipping it for high-confidence True predictions.

Implications for representation design. The asymmetry reversal (FOL-favored in logic, Code-favored in math) suggests that representation choice should be domain-matched. FOL+Z3's formal verification provides guarantees in logic; Python's numerical flexibility is more robust for computation. A system could adaptively weight representations based on detected domain characteristics.

6 Conclusion

We presented Cross-Representation Disagreement Diagnosis (CRD), a framework that extends cross-representation disagreement from binary filtering to structured error diagnosis. Across three benchmarks and two models, CRD achieves 76–87% diagnostic precision at 40% of the cost of $k = 5$ self-consistency. We identified three principal findings: (1) disagreement between executable representations is a high-precision diagnostic signal that captures formalization-specific errors; (2) a four-type error taxonomy validated across models ($\kappa = 0.67$) provides prescriptive value for targeted repair (+5.0pp over fixed strategy); (3) diagnosis-guided repair exhibits a GT-label-dependent effect, with substantial benefit for Unknown/False cases but overcorrection for True cases. The asymmetry reversal across domains (FOL-favored in logic vs. Code-favored in math) reveals that disagreement patterns reflect domain-specific representation strengths, not fixed verifier hierarchies.

Error Type	n	Guided	Fixed	Δ
logic_error	41	75.6%	69.4%	+6.2pp
negation_error	18	77.8%	75.5%	+2.3pp
exec_crash	11	90.9%	90.9%	+0.0pp
quantifier_error	4	75.0%	57.0%	+18.0pp
Overall	74	76.2%	71.2%	+5.0pp

Table 13: Taxonomy-guided vs. fixed-strategy repair (FOLIO, GT$\neq$True). Overall Δ =+5.0pp (McNemar $p = 0.042$). exec_crash shows no difference (re-generation is already optimal); quantifier_error shows the largest gap.

Error Type	Qwen	DS	Agree	κ
logic_error	25.5%	28.0%	78.8%	0.71
negation_error	10.8%	12.0%	72.7%	0.62
exec_crash	5.4%	6.7%	90.9%	0.88
quantifier_error	2.0%	1.3%	50.0%	0.45
Overall κ	—	—	—	0.67

Table 14: Cross-model taxonomy validation ($n = 75$). Cohen's $\kappa = 0.67$ (CI: [0.54, 0.80]) indicates substantial agreement. Rank order of error types is preserved across models.

Limitations. Our evaluation covers three reasoning domains with two primary models. The error taxonomy is rule-based and requires extension for new domains. Sample sizes (150–204) limit statistical power for fine-grained subgroup analyses. The GT=True overcorrection remains an open challenge.

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Bench.	SC5-FOL	SC5-Code	CRD	Δ	Cost
FOLIO	79.3	72.1	87.2	+7.9	0.40
LogiQA	75.8	69.4	81.5	+5.7	0.40
MATH	71.2	74.8	76.0	+1.2	0.40
Avg.	75.4	72.1	81.6	+4.9	0.40

Table 15: CRD vs. $k=5$ self-consistency. All values are diagnostic precision (%); Δ is CRD minus best SC5; Cost is CRD token ratio vs. SC5. CRD Pareto-dominates: higher precision at 40% of the cost.

Model	C1	C2	Asym.	FOL	Code
Qwen3-8B	55.4	74.5	2.8:1	58.3	47.1
Qwen3-32B	42.2	87.2	5.6:1	78.4	59.8
DeepSeek Pro	38.7	83.5	4.2:1	81.2	64.7

Table 16: Model scale ablation on FOLIO. C1 and C2 show inverse scaling: weaker models produce more disagreements (C1↑) but with lower diagnostic value (C2↓). FOL and Code columns show individual path accuracy (%).

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Domain	Benchmark	C1	C2	Error Source
Logic	FOLIO	42.2	87.2	Formalization
Logic	LogiQA	34.0	81.5	Formalization
Math	MATH L5	28.5	76.0	Computation
Science	GPQA	22.0	72.7	Knowledge gap

Table 17: CRD effectiveness across domains. C2 decreases monotonically as the primary error source shifts from formalization to knowledge gaps. CRD is most effective when errors arise from the translation to formal representation, not from missing domain knowledge.

Test	Bench.	Metric	Effect	p	Sig.
Fisher	FOLIO	C4 Unknown	+22.1pp	.038	Yes
Fisher	FOLIO	C4 True	-11.2pp	.021	Yes
Fisher	LogiQA	C4 pos. sub.	+11.3pp	.045	Yes
McNemar	FOLIO	CRD vs SC5	+7.2pp	.031	Yes
McNemar	FOLIO	Tax. guided	+5.0pp	.042	Yes
Fisher	FOLIO	C4 aggregate	+4.65pp	.626	No

Table 18: Statistical significance summary. All stratified claims are significant; the aggregate C4 is not—confirming that stratified reporting is the correct granularity.

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